Youssef Naciri Zerta

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Research Vision

I build toward interpretable, identifiable representations across the modalities of life — what some call *world models of biology*: representations that recover mechanism rather than merely fit data, and earn trust by surviving identifiability and false-discovery control instead of being read off on faith. My work is a single program seen from successive layers of that model — generating the data at the bench, grounding it in known structure and prior knowledge, learning the latent state space, integrating multi-omics into predictive readouts, and auditing whether the learned representations are real. Each project below is a layer of that one stack; my distinctive angle is working the entire loop, and holding every representation to the standard of mechanism that survives scrutiny.

Research Experience

Lead Developer — AnnNet 2025–Present

EMBL-EBI, Saez Lab — *the structure & prior-knowledge layer — the known mechanism a world model is grounded in, not just learned*

- Architecting the core engine of AnnNet, a standardized Python library for high-performance biological network science and graph analytics — the prior-knowledge substrate mechanistic, interpretable models are built on.
- Building APIs for topological representation and graph-based inference used by groups at EMBL-EBI and Imperial College London; standardized cross-institutional data structures keep signaling-network models reproducible at scale.
- Growing AnnNet into SysBioVerse, an open, interoperable, *scverse*-style ecosystem for the structural layer of a world model of biology.

Research Collaborator — Single-Cell Representation Learning 2025

Helmholtz Munich, Lücken Lab — *the latent-state layer — learning the state space a world model operates in, and testing whether that representation transfers across systems*

- Co-developed an architecture combining contrastive learning and multi-head attention to learn patient-level embeddings from single-cell transcriptomics.
- Built a scalable latent representation reaching state-of-the-art results on disease-stratification benchmarks, and studied its transfer behavior as a probe of representation quality.
- Preprint: *Great patients embed alike: contrastive learning for sample representation from single-cell data*. Stack: PyTorch, GNNs, single-cell genomics, contrastive and representation learning.

Research Collaborator — AI-Powered Biomarker Discovery 2024–Present

MD Anderson Cancer Center & KAUST — *the multi-omics readout layer — from molecular state to cancer outcome to candidate biomarker, validated statistically*

- Built an end-to-end machine-learning pipeline for cancer biomarker identification from multi-omics data — raw genomics through feature engineering, model training, and statistical validation.
- Identified novel biomarker candidates across multiple cancer types and produced publication-ready analyses. Stack: PyTorch, scikit-learn, dimensionality reduction, survival analysis, pathway enrichment.

Undergraduate Researcher — Multiplex Bacterial Interaction Networks 2024–Present

Université Euromed de Fès (UEMF) — *the organism-in-environment layer — a representation of an organism in its ecological context across 12 modalities*

- Designed a trait-resolved multiplex interaction study across 60 soil bacterial strains, generating 12 directed interaction layers from large-scale pairwise assays.
- Built a multilayer network-analysis pipeline (community detection, motif analysis, node-role characterization) over a 3D interaction tensor, and derived a 4-dimensional ecological embedding classifying microbial strategies across antagonistic, metabolic, and morphogenetic modes.

Causal Identifiability of Sparse-Autoencoder Features in Foundation Models 2026

Solo author, under submission — *the auditing & identifiability layer — testing whether a world model's learned features are real, not read on faith*

- A causally-grounded test of whether the interpretability "features" read out of a foundation model are trustworthy, using saturation-mutagenesis MPRA for per-base-pair causal ground truth.
- Two-sided result: a real aggregate causal signal survives a full control suite, yet the per-feature SAE decomposition is non-identifiable — causal features barely reproduce across retrains, a second SAE family recovers a near-disjoint

causal set from identical activations (cross-family Jaccard 0.01), and a single supervised direction stably decodes the effect. The failure is the decomposition, not the model.

- Generalizes across two SAE families (TopK, Gated) and two architectures (HyenaDNA, Caduceus). Implication: individual model features must not be read as mechanism without seed-stability and false-discovery control — directly relevant to model auditing and AI security.

Education

State Engineer (Diplôme d'Ingénieur d'État) — Biotechnology Expected June 2027

Université Euromed de Fès (UEMF), Morocco — *wet-lab-trained biotechnology engineer*

Coursework: Metabolic Engineering, Recombinant DNA Technology, Advanced Microbiology, Viral Genetics, Optimization, Probability & Statistics.

Technical Skills

Interpretability & Identifiability: Sparse Autoencoders (SAEs), Mechanistic Interpretability, Feature Identifiability, False-Discovery Control, Causal & Mechanistic Modeling.

Machine Learning: Representation Learning, Contrastive Learning, Graph Neural Networks (GNNs), Foundation Models, Symbolic Regression.

Computational Biology: Single-Cell Genomics, Multi-Omics Integration, Network & Mechanistic Modeling, Survival Analysis, Pathway Enrichment.

Wet Lab: Recombinant DNA Technology (RDT), Genome Engineering, Molecular Cloning, High-Throughput Bacterial Culture.

Engineering & Math: Python (PyTorch, scikit-learn), C/C++, R, Bash, Docker, Google Cloud Platform (GCP), API & Library Design; Optimization, Probability & Statistics, Dynamical Systems.

Languages: English & French (Fluent), Arabic & Tamazight (Native), Spanish (Basic).